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Perception of ambiguous rhoticity in Glasgow

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ABSTRACT

Relatively little research has been conducted on the effect of hearing an unfamiliar native English accent. This paper tests listeners with varying levels of familiarity with the Glaswegian linguistic environment, presenting them with naturalistic minimal pairs such as *hut/hurt* – produced by speakers raised in Glasgow – in two-alternative-forced-choice tasks. The results of Experiment 1 show a benefit of long-term familiarity in discriminating minimal pairs with derhoticised /r/, a phonetically eroded form of postvocalic /r/ in working class Glaswegian. Native Glaswegian listeners displayed high sensitivity to difference (*d'*), and low response bias (*c*) towards hearing either rhotic or non-rhotic words, indicating accurate perception. Unfamiliar listeners were less sensitive to stimulus difference, and were biased towards hearing plain vowels, demonstrating their unfamiliarity with Glaswegian /r/. Non-rhotic English listeners with a moderate level of experience with Glaswegian showed an effect of 'perceptual hypercorrection', i.e. over-reporting /r/ presence. Experiment 2 found that, following a short period of exposure, English listeners with very little experience with Glaswegian also started to show hypercorrection, suggesting rapid adaptation to novel phonetic detail. These results may be explained by some general principles underlying exemplar and hybrid theories, and contribute to the ongoing research into the complex nature of Scottish /r/.

1. Introduction

Challenges to speech perception may include a lack of familiarity with certain characteristics of the speech signal, or the presence of multiple speakers. If the listener is very familiar with the speaker's accent, they should find it relatively easy to correctly decode the intended message. However, the additional challenge presented by hearing an accent with unfamiliar phonetic features can adversely affect the listener's ability to accurately process speech. This paper directly deals with this the challenge of accent familiarity in a detailed measurement of listeners' perception of one particular fine-grained, dialect-specific phonetic realisation of a phonological contrast, in order to assess the impact of these challenges to accurate speech perception.

This study makes use of the phenomenon of rhoticity to measure how listeners' perception of fine phonetic detail can vary across certain factors. Rhoticity is an ideal testing ground for such a study, given the range of rhotic variants showing sociolinguistic stratification, in English in general (e.g. Labov, 1986), and in Scottish English in particular (e.g. Lawson et al., 2014; Lawson et al., 2018; Sóskuthy and Stuart-Smith, 2020). This paper presents the results of two listening experiments which both investigate the perception of derhoticised /r/ in Glasgow, by listeners with different levels of experience with the feature. The results are then presented in a discussion of the role of experience – both long-term familiarity and short-term adaptation – in listeners' ability to perceive difficult phonetic contrasts in a non-standard accent of English.

1.1. Rhoticity in Scotland

Scottish speech is in general rhotic, while non-rhoticity is the overwhelming norm in England (Wells, 1982). Around 70–80% of Scotland's population lives in the heavily-populated Central Belt, with the largest concentration found in the west, in Glasgow (Scotland's largest city), which has a complex linguistic environment. Despite possibly having a greater sense of Scottish identity than the middle class (MC) community, some features produced by working class (WC) Glaswegian speakers are changing towards realisations found primarily in working class London speech, with Stuart-Smith et al. (2007, 2013) suggesting this may be due the influence of the UK-wide broadcast media. Taps and trills, once a strong feature in Scotland (Grant, 1913; Johnston, 1997; Stuart-Smith

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et al., 2014), are no longer as commonplace in any Scottish variety. Nevertheless, middle class Glaswegian speech is maintaining more traditionally Scottish features than working class speech, despite having more opportunities for contact with Anglo-English speakers and weaker network ties (Stuart-Smith et al., 2007). One reason for this difference is that the existence of strong network ties in working class communities (e.g. Milroy and Gordon, 2008) helps to spread changes – including weaker acoustic rhoticity – in a much faster way than linguistic changes happen in middle class communities.

A study by MacFarlane and Stuart-Smith (2012) showed the importance of indexical cues in Scottish /r/ variants, with Glaswegian listeners (mainly middle class) categorising approximant [ɹ] in postvocalic position as indicative of middle class speech, and tapped [ɾ] in the same position as indicating working class speech. However, the authors found that social judgements of [ɹ]∼[ɾ] variation in onset /r/ did not follow a similar pattern as judgements of postvocalic /r/ (as observed by Stuart-Smith, 2003), being categorised at chance level. Thus it seems that more indexical information is contained within postvocalic /r/ than in other positions – indeed, Lawson and Stuart-Smith (2021) found that utterance-final position may be a key location for the performance of social variation in Glaswegian /r/. In an unpublished undergraduate study, Ashton (2011) found that listeners raised in Scotland's Central Belt would associate a strongly rhotic /r/ variant with middle class Edinburgh speech, and derhoticised and r-less realisations with both working class and Glaswegian speech. This result appears to lend support to MacFarlane and Stuart-Smith's (2012) finding of a strong social significance of different articulations of postvocalic /r/ in Scotland. Furthermore, in Scottish/English border towns, Watt et al. (2010) found that rhoticity/non-rhoticity was both stable within, and different between, the speech communities, indicating that /r/ may also have particularly strong social salience in speech perception (Trudgill, 1986). Further evidence of this comes from Sumner and Samuel's (2009) finding that when rhotic General American listeners heard examples of a New York non-rhotic accent, they experienced a clear and consistent processing cost associated with their lack of experience with the accent. These listeners were much less accurate when processing words like *baker* with a non-standard r-less pronunciation (bak[ə]), than they were at processing the standard r-ful pronunciation (bak[ɚ]). However, listeners who were raised in the r-less accent area of New York found it almost as easy to process r-less tokens as r-ful tokens. This result is relevant for the present study because it appears that the standard, dominant r-ful pronunciation facilitates processing for both groups of listeners, regardless of their linguistic experience.

Lennon et al. (2015) provide a useful comparison between the /r/ variants produced by middle class and working class speakers in Glasgow, conducting a phonetic analysis of the stimuli used in Experiment 1 of the present work. In line with the findings by Lawson and colleagues (e.g. Lawson et al., 2018) that young speakers in Glasgow show a high degree of social stratification in their rhotic variants, Lennon and colleagues found that the two middle class Glaswegian speakers in their data produced a strongly-rhotic schwar variant [ɚ] for postvocalic /r/, with the two working class speakers using a derhoticised, pharyngealised variant [Vˤ] for /r/. They found that the middle class stimuli had clear differences between the formant structures in minimal pairs such as *bud/bird*, *hut/hurt*, and *thud/third*, meaning that the minimal pairs were acoustically distinct, but the working class stimuli were much more similar in the formant structures for the same minimal pairs, such that there was very little difference between the formant trajectories in the V(+/r/) vocalic portion (Plug and Ogden, 2003; Stuart-Smith, 2007) between the minimal pair counterparts. This was especially the case in F3, which displayed differences which are said to be crucial for a clear perception of acoustic rhotic variants.

Although no articulatory data are discussed in the present work, it is interesting to note a potential contribution that this paper can make to the existing articulatory literature on Scottish /r/ variants, which are characterised by the same highly distinctive (and socially marked) acoustic patterns tested here. Lawson et al. (2011b) and Lawson et al. (2014) suggest a potential correlation between the auditory realisations of /r/ in both middle class and working class speakers, reproduced here as Fig. 1, and the same speakers' articulatory configurations, shown here as Fig. 2 (taken from tongue-spline measurements made in Articulate Assistant Advanced, Wrench, 2007).

These two graphs each show a clear split between the 'strong' and 'weak' /r/ productions of middle class and working class speakers in the Scottish Central Belt, and the same split can be seen across both graphs. Middle class speakers (diamond symbols in Fig. 1) have stronger auditory /r/ ratings (higher on the y-axis), corresponding to more front- and mid-bunched articulations for /r/, and working class speakers have much weaker auditory /r/ ratings, corresponding to more front- and tip-up (i.e. derhoticised) articulations for /r/.[1] The strong correlation between the patterns in the two analyses is confirmed by the authors. Interestingly, there appears to be slightly more class polarisation in the Western Central Belt auditory data (i.e. Greater Glasgow) than in the Eastern Central Belt auditory data (i.e. Edinburgh and surrounding areas). The pattern is perhaps not as notable in the articulatory data (Fig. 2), but companion perceptual experiments in the East and West Central Belt would shed more light on the perceptual correlates of these data. Experiment 1 (described in Section 2) is well-placed to test whether the phoneticians' auditory judgements of these articulatory configurations are replicated in the wider population, as it tests listeners on different socially-stratified realisations of Glaswegian /r/. In Section 2.7 these findings will be discussed in relation to the results of Experiment 1, described later in this paper.

1.2. Phonetic properties of postvocalic /r/

In most of the previous literature, F3 lowering towards F2 is said to be important for the perception of acoustic rhoticity. Indeed, Lindau (1985) suggests, using primarily English data, that a lowered third formant is an acoustic factor common to rhotic sounds. This appears to be the case for most approxi-

[1] Note that the term 'tip-up' used by Lawson and colleagues does not necessarily refer to a retroflex articulation for this sociolect, as the raised-tip gesture primarily occurs after voicing has ended, and is therefore more likely to be associated with a derhoticised variant (e.g. Lawson and Stuart-Smith, 2021).

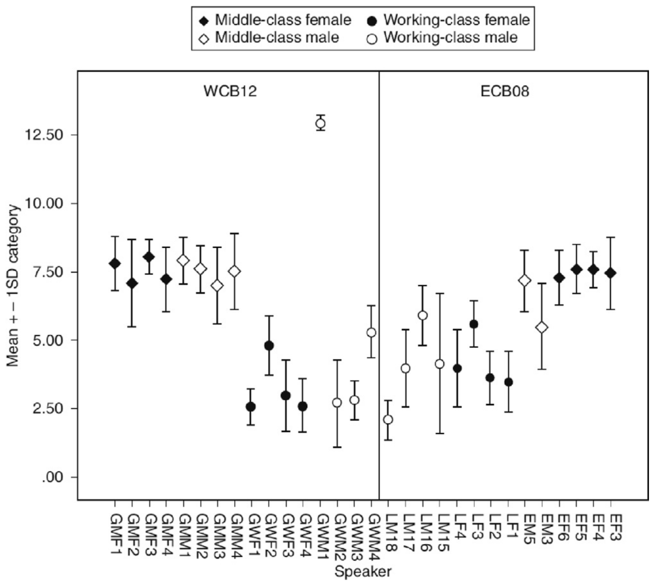


Fig. 1. Phoneticians' auditory ratings of /r/-strength in speakers in/near Glasgow (WCB) and Edinburgh (ECB). Individual speaker /r/-index score means ± one standard deviation. WCB12, n = 394, ECB08, n = 136 (from Lawson et al., 2014, 67).

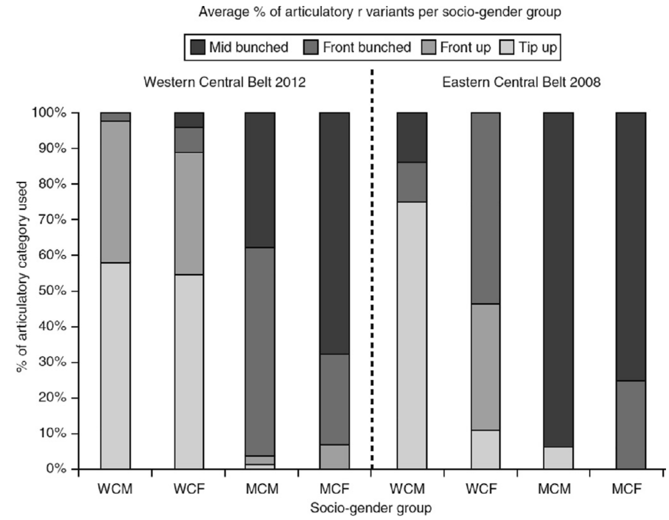


Fig. 2. Percentage of articulatory /r/ variants used by each socio-gender group in the western and eastern Central Belt. WCB12, n = 394, ECB08, n = 136. Shades from lightest to darkest represent TIP UP, FRONT UP, FRONT BUNCHED and MID BUNCHED configurations respectively (from Lawson et al., 2014, 70).

mant rhotics in English and other languages. However, a lowered F3 is not apparent in other types of rhotics; for example, voiced and voiceless uvular /r/ in Swedish, French, and German all have a high F3 (e.g. Maddieson and Ladefoged, 1996; Stuart-Smith et al., 2014).

In Standard Scottish English (SSE), a lowered F3 is indeed an indicator of strong acoustic rhoticity, as it corresponds to alveolar, retroflex, or more commonly a bunched approximant for coda /r/ in this variety (Lawson et al., 2014; Lennon et al., 2015; Jauriberry, 2021). However, this is not the case for all varieties in the Scottish linguistic environment. Stuart-Smith (2007) describes a weakened form of rhoticity in Scotland – derhoticised /r/ – as having a pharyngealized or uvularized quality, noting the slight rise and weakened amplitude of the third formant associated with the feature. Other papers make similar observations about raised or flat, vowel-like F3, which is a feature associated with working class speech in the Central Belt's urban areas (e.g. Lawson et al., 2010; Lawson et al., 2011a; Lawson et al., 2011b; Lawson et al., 2014; Purse and McGill, 2016; Sóskuthy and Stuart-Smith, 2020). Furthermore, articulatory studies using ultrasound tongue imaging show that Scottish derhoticised /r/ displays a delayed tongue tip raising gesture together with an early tongue root retraction gesture (Lawson et al., 2011b; Lawson and Stuart-Smith, 2021; Stuart-Smith et al., 2014), and so there is often a degree of constriction at the pharyngeal/uvular place of articulation. Because of this, Scottish derhoticised /r/ appears to share some acoustic and articulatory properties with uvular rhotics found in other languages.

Heselwood (2009) and Heselwood and Plug (2011) tested the effect of certain acoustic properties of /r/ in a residually rhotic variety in East Lancashire, Northern England, on the auditory judgements of strength of acoustic rhoticity by phonetically trained listeners. It was found that when F3 lowered towards F2, an auditory judgement of rhoticity was reported by the listeners. However, when F3 was attenuated or even removed from the same stimuli using acoustic manipulation techniques such as low pass filtering, an increase in the perceptual strength of the /r/ was reported by the listeners. Because of this finding, Heselwood and Plug (2011) suggest that the widely held assumption that a low F3 is a crucial acoustic and auditory correlate of rhoticity should be refined (Heselwood and Plug, 2011). Indeed, Heselwood et al. (2010) write that it is the location of formants in auditory space (which can be quantified through a psychoacoustic conversion such as Bark transformation) rather than acoustic space that best predicts perceptual judgements of strength of rhoticity. Furthermore, they suggest that the presence of a strong F3 close to the F2 frequency range may in fact inhibit the perception of acoustic rhoticity (Heselwood, 2009; Heselwood and Plug, 2011). This is because of a combination of effects: the closer F3 gets to F2, the stronger the perceptual peak is around the F2 frequency range – and the further away from F4 it is – resulting in a more complete auditory integration around F2 (Bladon, 1983; Chistovich and Lublinskaya, 1979; Hayward, 2014). If F3 is now removed entirely, this means that the strong perceptual peak around F2 is lowered even further away from F4, thus increasing the distance between perceptual peaks in the F2 and F4 ranges.

Turton and Lennon (2023) conducted an analysis of the same variety of Northern English as Heselwood and colleagues – in the town of Blackburn, specifically – finding that an increase in F3-F2 distance was the major indicator of weakening acoustic rhoticity in the variety over apparent time. They note that this change is not as clear cut as a simple acoustic weakening, with many young speakers performing a schwa-offglide in their ostensibly non-rhotic productions of words such as *sore*, but no such offglide in their minimal pair counterparts like *saw*. This indicates that speakers may conceptualise rhotics as different phonemic entities from plain vowels, even though they may show no signs of phonetic rhoticity in the expected acoustic terms. Indeed, the authors note that the youngest speakers in their corpus of Blackburn speech may be producing offglides as an intermediate variant of postvocalic /r/. This suggests that offglides may be one of the final stages in an imagined historical progression of derhoticisation at the community level, from trilled /r/ to plain vowel, in e.g. *'deer'*: [dir] → [diɾ] → [diɻ] → [diɹ] → [diə] → [dɪː][2] – the last of which is seen in less conservative forms such as Estuary English (also see Stuart-Smith et al., 2014, who describe one definition of derhoticisation as gradient phonetic lenition process from trill to complete /r/-loss). Auditory analysis and perceptual testing would help to determine whether listeners (native to Blackburn or otherwise) are sensitive enough to perceive these differences in fine phonetic detail as different phonemic categories.

A problem that may arise in acoustic analyses in some forms of rhoticity is inconsistent formant intensity, especially in field recordings, which is especially important for F3. This would make it difficult to determine which acoustic cues listeners are attending to when they perceive acoustic rhoticity. An example of this is found in the literature on Scottish English. In the derhoticised form of /r/ produced by some Glaswegian speakers, the relevant acoustic cues can be elusive. Stuart-Smith and colleagues (e.g. Stuart-Smith et al., 2014) found that derhoticised /r/ showed, variously, either rising or falling F3, which may depend upon the articulatory configuration of the front cavity in each case, and often they found that a drop in spectral intensity made the measurements hard to make. In such cases, perceptual cues may therefore be difficult to investigate without a fuller understanding of the articulatory variation which is present in Glaswegian /r/. It should also be borne in mind that acoustic rhoticity and /r/ might be present in the signal in ways not picked up by formant trackers, due to e.g. coarticulation in voiceless noise sources (e.g. Lawson et al., 2018).

1.3. Perception of acoustic rhoticity

We shall now examine some evidence about the effect that the articulatory configuration of different /r/ variants has on the perception of acoustic rhoticity, focusing on Scottish /r/ in particular. Delattre and Freeman (1968) describe many articulatorily distinct forms of the American English phoneme /ɹ/, but they report that in general it is said to be produced with either a bunched or a retroflex articulation, and that the auditory impression is the same for both articulations. Because of this, along with Lindau's (1985) claim that American speakers use all available articulatory methods that they can in order to produce a low F3 (meaning that similar acoustic profiles may result from different articulations), it seems reasonable to assume that retroflex and bunched /r/ are entirely auditorily equivalent. Indeed, Mielke et al. (2006) report that variation between bunched and retroflex articulations in American English is speaker-specific, so we may assume that, at least in American English, an individual can arbitrarily choose either articulation. However, Zhou et al. (2007) found a difference in the acoustic relationship between the fourth and fifth formants between bunched and retroflex /r/ articulations.

Work by Lawson and colleagues may suggest that there is some social salience attached to bunched articulation of /r/, as opposed to the retroflex articulation, despite the findings of Twist et al. (2007) which suggested that listeners could not reliably distinguish between them in American English. Using Ultrasound Tongue Imaging recordings, Lawson et al. (2014) reported that middle class Glasgow and Edinburgh adolescent speakers categorically produced /r/ with a bunched articulation, rather than retroflexion, which was only found to be produced by the working class speakers in the same study. Middle class Scottish speakers have become increasingly rhotic (Lawson et al., 2011b; Lawson et al., 2014; Lawson et al., 2018; Lennon, 2012), with the loss of the DRESS lexical set in a subset of NURSE words leading to a merger (Lawson et al., 2013). This phonological interpretation, accompanied by the fact that working class speech is continuing to undergo derhoticisation (e.g. Lawson et al., 2011b; Lawson et al., 2014; Lawson et al., 2018; Stuart-Smith et al., 2014), may underline the social significance of a perceptually strong rhotic variant.

One study which focuses specifically on the perception of derhoticised /r/ is an unpublished dissertation by Bond (2013), which found that in the absence of strong perceptual differences in working class Glaswegian /r/, like a perceptually-salient spectral peak at F2/F3 (found in middle class /r/), listeners appear to use more subtle phonetic detail to make their decisions about the identity of ambiguous words. In his study, Bond presented working class Glaswegian listeners with acoustically manipulated (i.e. cross-spliced) recordings of working class Glaswegian tokens *had* and *hard* with derhoticised /r/, finding that they attended to the quality of both the preceding vowel and the following plosive in order to correctly identify the word. Interestingly, Bond notes that one of the participants in his perceptual test reported hearing all 14 of the manipulated stimuli as /r/-less, i.e. as 'had' rather than 'hard', even when the speaker intended to produce e.g. *hard* (Bond, 2013, 53). One possible explanation Bond puts forward for this is that the participant may have been attending to only the apical articulation (e.g. [d]) rather than any of the other acoustic cues in the stimuli. Perhaps more interestingly however, Bond hypothesises that 'this participant uses the presence [in the vocalic portion] of a rising F2 transition to the following coronal plosive as the main cue for identifying /r/-less words'. In fact, Bond's experimental stimuli were manipulated so that the F2 frequency of the vocalic portion in each stimulus was constant, meaning that all vocalic portions had the characteristics of the derhoticised /r/-ful tokens, as opposed to the /r/-less tokens, which often had a varying F2 frequency.

At this stage an important clarification should be made concerning the interaction between listener experience and the nature of the phonetic parameter across which the minimal pair operates. The fact that a certain listener group may experience difficulty with ambiguous minimal pairs across the phonetic parameter of gradient derhoticised /r/, i.e. *hut/hurt*, does not entail that the same group of listeners have difficulty in distinguishing minimal pairs across the parameter of, say, voice onset time (VOT), for example *pie/buy*. Instead, the fact that the cues to determining contrastive VOT are commonly encountered by speakers of English, whereas cues to derhoticisation are not, is key to the present investigation.

1.4. Perceptual adaptation

In the event that Experiment 1 demonstrates a performance difference related to listener experience when discriminating

[2] Thanks are due to an anonymous reviewer for this useful observation of an imagined historical progression towards non-rhoticity.

ambiguous phonetic detail in derhoticised /r/ – thus motivating a test of listener adaptation in Experiment 2 – it is important that we now consider previous research into perceptual adaptation to novel or unfamiliar accents. In a lexical decision task using an Exposure-Test design, Maye et al. (2008) found that American English listeners who were given the opportunity to learn a novel phonetic difference showed rapid adaptation to a talker with an acoustically-manipulated vowel set (e.g. *west* became *wast*). Importantly, the authors report that the listeners did not simply relax their phonemic categories to accept any direction of vowel shift, as they only adapted their expectations to match the vowel qualities specific to the talker to which they were exposed. Best et al. (2015) reported similar findings, with Australian listeners benefiting from exposure to London and Yorkshire English, compared to a control group who did not receive prior exposure to these unfamiliar varieties. However, in a related study with an increased number of unfamiliar accent groups, Shaw et al. (2018) did not replicate this effect of rapid adaptation across all accent exposure groups, leading the authors to speculate that their Australian listeners' perceptual flexibility may have been more sensitive to prior experience with (and/or attitudes about) particular accents than was accuracy in the vowel categorisation tasks they completed. For example, they note that listeners did not display a benefit for adaptation following exposure to New Zealand vowels (compared to London vowels), with the authors highlighting the fact that New Zealand speech is subject to negative regard in Australia.

Supporting their hypothesis that prior experience likely has an effect, Adank et al. (2009) found positive effects of long-term exposure when Glaswegian listeners heard speech produced in Southern Standard British English (SSBE, the dominant variety in UK broadcast media), but the fact that SSBE listeners incurred a processing cost when hearing the Glaswegian accent could reflect the fact that, as a highly regional variety in the UK, Glaswegian is much less familiar to SSBE listeners than SSBE speech is to Glaswegians. Although Adank et al. (2009) describe an experiment testing long-term exposure, it provides useful background for studies reporting positive findings when testing the benefit of prior experience in adaptation to non-native speech, with e.g. Kim et al. (2020) finding that listeners are sensitive to the informativeness of certain phonetic cues.

In order to facilitate a theoretical framework against which to compare the results of this paper, it is useful to categorise work on speech perception into different approaches, most of which have arisen over the last few decades. Firstly, studies which align with exemplar theory (sometimes termed episodic theory) arose from influential work reported through the 1990s (e.g. Mullennix et al., 1989; Mullennix and Pisoni, 1990; Nygaard et al., 1994; Pierrehumbert, 2001), and is characterised by the assumption that every perceived occurrence of speech is recorded and stored by the listener, and over time categories of occurrences emerge from the way the information is stored. The listener matches novel incoming features (phonetic cues, phonemes, words, etc.) against their set of stored representations, with this matching process enabling the listener to make probabilistic judgements about which phoneme (or other feature) was produced. In opposition to earlier, abstractionist theories in which phonetic variation, such as varying speaker identities, is stripped away, revealing the higher-level categories (Chomsky and Halle, 1968), Johnson (1997) wrote that listeners may be able to perform speaker normalisation if they use their stored exemplars of speech to evaluate words produced by similar talkers (see also Nygaard and Pisoni, 1998). Indeed, coping with accent variation could be seen as an extreme form of speaker normalisation, with the present speaker's acoustic signal being compared to all other signals from other speakers (e.g. Hay et al., 2006; Evans and Iverson, 2004; Evans and Iverson, 2007).

More recently, exemplar positions have developed into more 'hybrid' approaches, allowing for both the storage of exemplars as well as more abstract categories as described by traditional, abstractionist models (e.g. Pierrehumbert, 2002; Pierrehumbert, 2006). Goldinger (2007) revises the pure exemplar approach of Goldinger (1996, 1998) into one that is hybrid, assuming that abstract knowledge is imposed upon each encountered stimulus, meaning that each stored exemplar is a product of perceptual input combined with prior knowledge. Hawkins and Smith (2001) proposed a model which incorporates rich, hierarchical structures incorporating prosodic and grammatical information, which improves the process of pattern-matching between signal and memories (Smith, 2015).

Finally, it is also possible to consider results of speech perception experiments – in particular those which involve an element of learning – under a theoretical framework using Bayesian inference, which is a relatively recent direction of research (Norris et al., 2016; Norris and McQueen, 2008; Feldman et al., 2009). One application of the Bayesian approach is Kleinschmidt and Jaeger's *ideal adapter* model (Kleinschmidt and Jaeger, 2015), whereby the listener has a set of prior beliefs about linguistic patterns, which are updated when new information arises, with the theoretical listener overcoming the problem of lack of invariance by conditioning speech perception on socio-indexical cues. Under this framework, Kleinschmidt et al. (2018) examine the relationship between the social and linguistic aspects of speech perception, claiming that there is a strong link between them, and they emphasise that speech perception seems to be sensitive to talkers' social identity, with listeners interpreting acoustic cues differently based on a range of perceived socio-indexical features. They conclude that there may be a deep, yet largely unexplored connection between sociolinguistics and psycholinguistics, which is supported by their results indicating similar processes of statistical knowledge that underpin social inferences and robust speech perception in the face of talker variability.

With this theoretical background as a reference point, we shall consider how the results presented in this paper might speak to the larger literature on speech perception, while making contributions to the ever-growing body of work on rhoticity in Scotland.

1.5. Research Questions

This paper describes two speech perception experiments, both investigating how listeners perceive contrastive and acoustically variable rhoticity in Glasgow English. The research questions are:

RQ1: 'What is the role of experience in the perception of coda /r/ versus its absence in Glaswegian English?' and;

RQ2: 'How does listener experience relate to the effect of exposure on the perception of a non-standard phonetic contrast?'

Experiment 1 addresses the first of these research questions by examining the role of long term listener experience in the perception of this contrast. Three groups of listeners with different levels of familiarity with Glaswegian take part in a speech perception experiment with a simple two-alternative-forced-choice task (2AFC) design. Experiment 2 then tests the same three group demographics in another two-alternative-forced-choice task, but adds the dimension of short-term exposure, in a Pretest-Exposure-Posttest experimental design, in order to address the second research question. The aim of this design is to determine whether listeners can learn the fine phonetic detail in derhoticised /r/ – and apply it to the perception of contrastive rhoticity – over the short duration of an experimental setting.

2. Experiment 1

The purpose of Experiment 1 was to assess the effect of long-term familiarity with derhoticised /r/, addressing *RQ1: 'What is the role of experience in the perception of coda /r/ versus its absence in Glaswegian English?'*.

2.1. Participants

The listener groups in this experiment were designed so that participants represented a range of experience with derhoticised /r/, but were still all native monolingual speakers of English. Therefore there were three accent groups, each with a different level of experience with the Glaswegian linguistic environment. There were 62 subjects (46 female) in these groups, with none reporting any hearing or language difficulties:

- **Glasgow** (n = 21): Raised in Glasgow, living in Glasgow;
- **Intermediate** (n = 21): Raised in England, living in Glasgow (mean residence = 3.6 years);
- **Cambridge** (n = 20): Raised in South East England (little/no experience of Glaswegian).

Participants were recruited from the student cohorts at the University of Glasgow (Glasgow and Intermediate groups), and the University of Cambridge (Cambridge group), and were paid a small fee for participation.

Despite the strong social salience of /r/ in Scotland, especially in the Central Belt, the socioeconomic status of the listeners was not controlled for in this experiment – this is an especially important consideration for the Glasgow listener group. Since the listeners were almost all recruited through university subject pools or advertisements on university noticeboards in both Glasgow and Cambridge, it is likely that most listeners will have been closer to the middle-class end of the socioeconomic spectrum. In addition, it is also the case that students attending the University of Cambridge are more likely to have attended private schooling than English students attending the University of Glasgow, which means that the demographics of the Intermediate and Cambridge groups may not be equivalent. These points will be considered in future research on this topic.

2.2. Materials

Experimental stimuli were the six word pairs shown in Table 1, which were extracted from recordings of four native Glaswegian males aged between 22–25 years. The speakers were recorded in a sound-attenuated booth, using lightweight Beyerdynamic headset condenser microphones at a sampling rate of 44.1 kHz. Each pair of speakers took part in a collaborative word-finding task: this meant that speech was as naturalistic as possible, while still ensuring that the desired set of target words was produced by each speaker. The speakers were two middle class childhood friends, and two working class childhood friends, recorded in separate pairs. The excision of the stimuli from these recordings of connected speech did not lead to any noticeable artefacts.

There were 144 word tokens in total (12 words x 4 speakers x 3 repetitions), balanced evenly across speaker and class (MC & WC). These stimuli were also used for the acoustic analysis conducted by Lennon et al. (2015), who found large acoustic differences between the middle class and working class tokens in their V(+/r/) portions in most environments, but only very small differences between the working class speakers' minimal pair counterparts in words with /ʌ/ vowels, such as *bud* vs. *bird*.[3]

The relative similarity between *hut/hurt* words produced by the working class speakers strongly suggests that there exists the potential for misperception of words containing derhoticised /r/ in the phonetic environment /ʌrC/. There is also anecdotal evidence of complete misperception of derhoticised /r/ in real world contexts, mainly by non-Scottish listeners to Scottish speakers, in television, radio, and podcasts. Fig. 3 shows the time-normalised tracks of the first five formants for the V(+/r/) portion of each word type in these stimuli. The vocalic portion of the lexically rhotic words was longer in duration than the lexically non-rhotic words in each minimal pair, potentially contributing to a greater percept of acoustic rhoticity by the listeners. This was the case for both the MC speakers, who produced strongly-rhotic schwar-type variants (e.g. Lawson et al., 2014; McAllister Byun and Tiede, 2017; Turton and Lennon, 2023) and the WC speakers, who produced weak, derhoticised variants (e.g. Stuart-Smith, 2007), nevertheless, normalised time is a useful way to visualise formant differences between tokens.

2.3. Procedure

Stimuli were presented in the perceptual testing software DMDX (version 3.2.5.1, Forster and Forster, 2003) on a Windows laptop computer, and participants wore over-ear head-

[3] The exception to this similarity is F2 in panel (b) of Fig. 3 (reproduced from Lennon, Smith, & Stuart-Smith, 2015), which shows the vocalic portion in words like *bud* and *hut*, produced by working class speakers. Since the vowel in these words was always followed by a coronal consonant (either /t/ or /d/ – see Table 1), the raising of F2 is associated with the formant transition into the consonant. Of course, this potentially constitutes a cue that listeners might be able to use to distinguish between the minimal pairs, but it was thought to be subtle enough that the selected recordings would be suitable as stimuli. This lack of absolute control of formant transitions highlights one of the downfalls of using naturalistic, non-manipulated speech in perception experiments (although see Section 3.2 for a description of how the manipulated stimuli in Experiment 2 partly address this).

Table 1
Minimal pairs used in Experiment 1.

/i/	/ʌ/
bead / beard	*bud / bird*
feed / feared	*hut / hurt*
weed / weird	*thud / third*

phones. After reading instructions, participants were asked to choose out of a pair of options, presented in orthographic form on the computer screen, around 1 metre in front of them. After a short practice session, participants heard 144 trials of individual tokens from the minimal pairs such as *bead/beard*, *hut/hurt*, randomised together.

The procedure for each trial was as follows: first, a blank screen was displayed, with the audio stimulus (e.g. a speaker producing the word *'hurt'*) played after a delay of 1 s. When the stimulus ended the two orthographic response options appeared on screen (e.g. HUT and HURT), at which point the participant would indicate which word they thought they heard by pressing a key labelled either 'L' or 'R' (for 'left' and 'right') on a keyboard. Responses were limited to 2500 ms, calculated from the moment the orthographic options appeared on the screen. At the end of the 2500 ms, the words would automatically disappear, and the trial would end. Following the 2AFC task, the participants completed a rating task which was not analysed as part of the current study.

It should be noted that because of its relative simplicity, participants were fully aware of the task at hand: choosing between a lexically rhotic and a lexically non-rhotic interpretation of each stimulus. This could have led to the participants attending particularly closely to the V(+/r/) portion, possibly more than under more naturalistic conditions.

2.4. Signal Detection Analysis

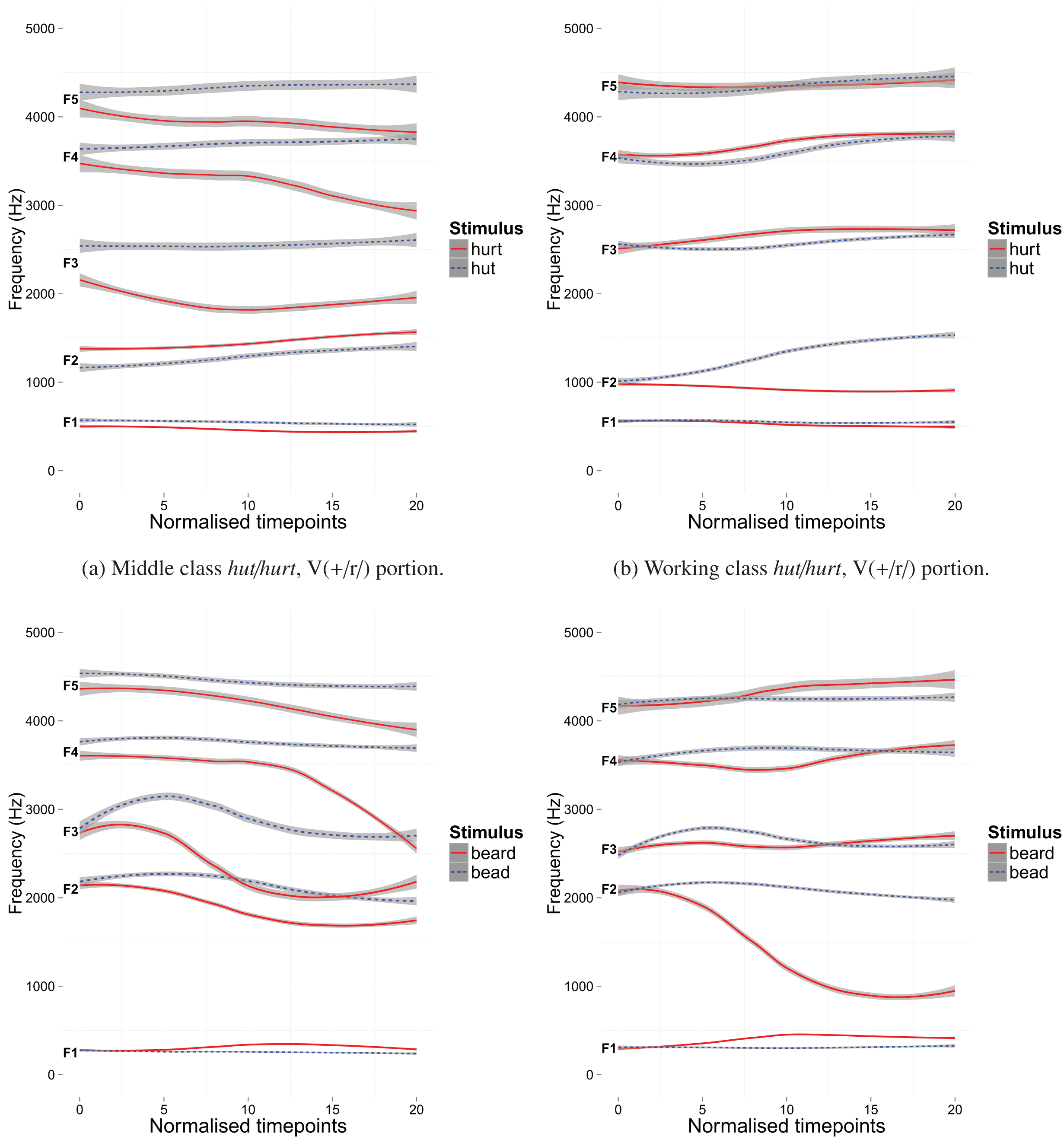
An analysis using Signal Detection Theory (SDT) was used in this study, enabling a more informative look at the patterning of responses than is afforded by analyses of response times or the percentage of correct responses (Macmillan and Creelman, 2005). In an SDT analysis, responses to a 2AFC task are classified in a 4-way response matrix which allows the researcher to assess the level of accuracy to *each type* of stimulus – in this case /r/-ful or /r/-less – meaning that bias to one response or the other can be calculated. Using the same matrix, sensitivity to stimulus difference can also be quantified. It was decided – because the /r/ was closest to being the target in this study – that the member of each minimal pair which orthographically had an <r> present, e.g. *hurt*, *beard*, etc., would be treated as the object that participants were 'aiming for' in the 2AFC. In other words, if a listener heard '*hurt*' and responded by selecting HURT (the *correct* response), this would be classified as a 'Hit', and if, for the same stimulus, the listener selected HUT (the *incorrect* response), this would be treated as a 'Miss'. To complete the 4-way matrix, if a listener heard the /r/-less member of the minimal pair, e.g. *hut*, and responded by selecting HUT (the *correct* response), this was classified as a 'Correct Rejection', and if for the same stimulus they selected HURT (the *incorrect* response), this was classified as a 'False Alarm'. The classification of these responses is summarised in Table 2, and the rates of each of these for each subject was calculated by aggregating each 'Hit', 'Miss', etc. on the level of the subject, as opposed to the less-common approach to aggregating by-item (Rabe et al., 2021). *d'* (sensitivity) and *c* (response bias) values were then calculated using the formulae provided in Macmillan and Creelman (2005, 21).

2.5. Statistical analysis

Despite the appearance of stimuli involving minimal pair contrasts such as *bead/beard* in Experiment 1, the decision was taken to remove the /i/ stimuli from the dataset, meaning that the factor of Vowel (levels = /i/ and /ʌ/) was not modelled, as it is not fruitful. The extremely high *d'* values for /i/ stimuli shown in Fig. 4, together with the relative lack of bias – seen in the proximity of *c* to 0 in Fig. 5 – indicate that listeners perform at ceiling, showing practically no variation in their perception of words with an /ir/ vocalic period. Accuracy levels were correspondingly high, with Glasgow listeners responding at 99.5% accuracy, Intermediate listeners at 99.1%, and Cambridge listeners at 98.4%. Therefore, in Experiment 1 we will be dealing with a subset of the data which includes only the /ʌ/ minimal pairs such as *bud/bird*, but still includes all levels of both Listener Group and Speaker Class factors (predictors listed in Table 3).

We now move on to a description of the modelling approach. Linear Mixed Effects Models were run separately on both sensitivity to difference (*d'*) and response bias (*c*), using the *lme4* package (Bates et al., 2015) in R (R Core Team, 2023) and RStudio (RStudio Team, 2022). Each model was built after subsetting the data to exclude the /i/ stimuli. The *d'* and *c* models were built using a fully-saturated approach to designing effects structures in LMER modelling, which involves including all possible effects in the model which are justified by the experimental design, then reducing where appropriate. It is common to include a random effect of word in modelling experiments of this type, but because the method of calculating *d'* and *c* (described in 2.4) makes use of values aggregated across the minimal pairs, it was neither possible nor necessary to include random effects relating to the individual words. Furthermore, it was also not possible to include a random slope of Class on Subject in the models described here (Barr et al., 2013), because there was an insufficient number of observations per factor level to do so. This was also a result of the aggregation of *d'* and *c* values across factors, which effectively created a single observation for each Class/Subject cell (i.e. one MC observation and one WC observation per subject). Indeed, Barr (2013) notes that in calculations with only a single observation per unit per cell, random slope variance cannot be distinguished from random error variance, meaning that random slopes need not be included. Thus, the only random effect included in the models was the random intercept of Subject.

Next, the *step()* function from the *lmerTest* package (Kuznetsova et al., 2017), which automates the process of backward-elimination of terms, was used in order to determine which factors were non-significant and could be removed, but all factors were found to contribute to each of the models. The predictors – with treatment-coded variables – for both the *d'* and *c* models are shown in Table 3.



(a) Middle class *hut/hurt*, V(+/r/) portion. (b) Working class *hut/hurt*, V(+/r/) portion.

(c) Middle class *bead/beard*, V(+/r/) portion. (d) Working class *bead/beard*, V(+/r/) portion.

Fig. 3. Formant tracks F1-F5 for all stimuli, averaged across Class and Vowel (graph reproduced from Lennon et al., 2015). Left panels: MC speakers; Right panels: WC speakers. Top panels: words with /ʌ(r)/; bottom panels: words with /i(r)/. /r/ stimuli are represented by solid lines, /r/-less by dotted lines. E.g. *'hut'* represents all *bud, hut, thud* stimuli. Plotted in R using *ggplot2*'s *stat_smooth()* function (Wickham, 2016) to draw formant tracks, after manual pre-smoothing in the Python-based program Formant Editor (Sóskuthy, 2014). Shaded ribbons: 95% C.I.

Table 2
Classification of 2AFC responses for signal detection analysis. *hurt* represents all stimuli with a postvocalic /r/ (*hurt*, *bird*, *third*), and *hut* represents all stimuli without a postvocalic /r/ (*hut*, *bud*, *thud*). Similarly, HURT represents all response options with a postvocalic /r/ and HUT represents all response options without a postvocalic /r/.

	HURT	HUT
hurt	Hit	Miss
hut	False Alarm	Correct Rejection

2.6. Results

2.6.1. Sensitivity *d'*

Since *d'* is a measure of sensitivity to difference between stimuli, a higher *d'* in Fig. 6 and in the Mean *d'* column in Table 4 represents better performance by the listener groups when deciding between minimal pairs in Experiment 1. It is clear from the *d'* values in Fig. 6 that listeners are very sensitive to the dif-

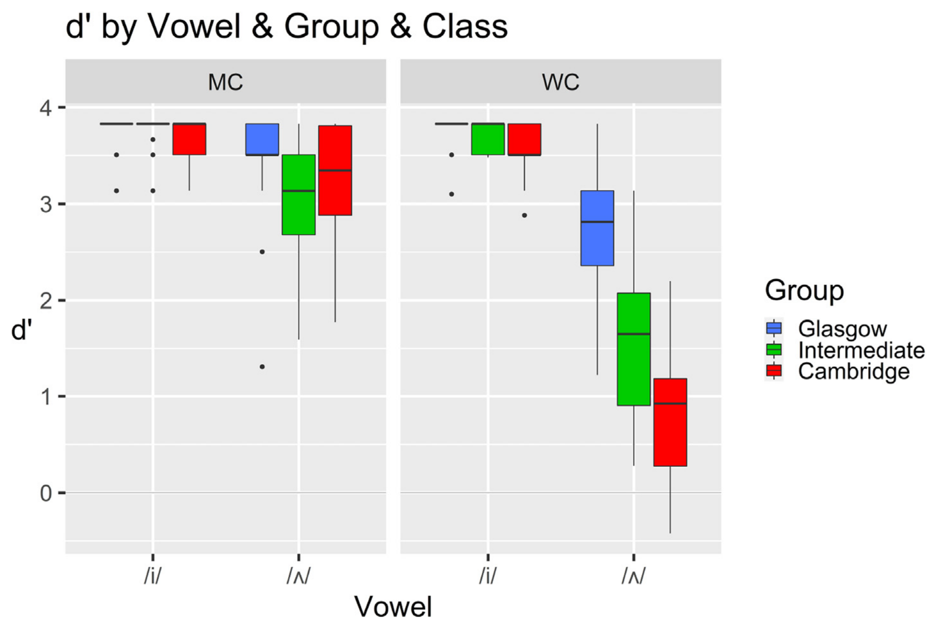


Fig. 4. Experiment 1 *d'* by Vowel, Group, & Class.

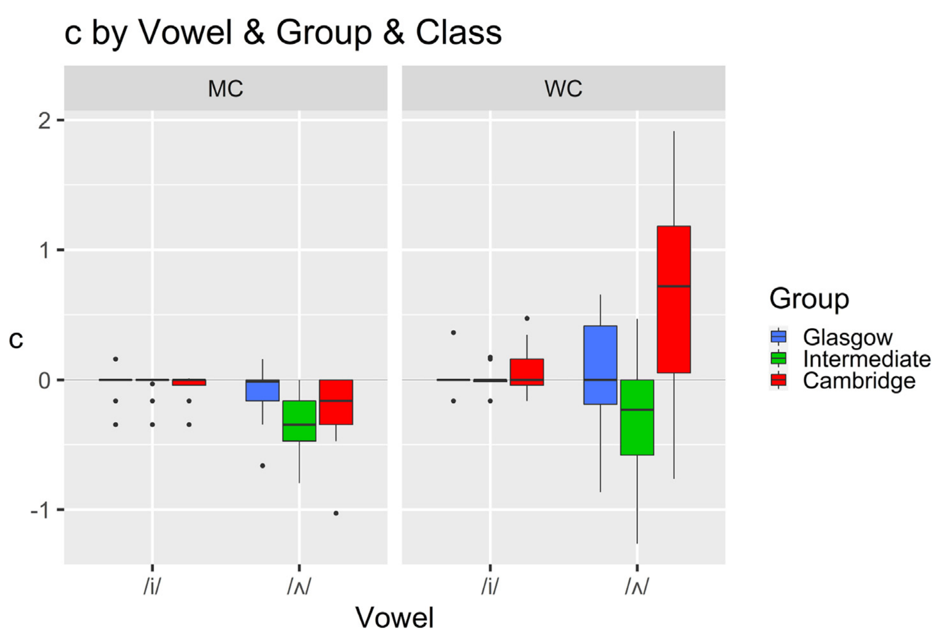


Fig. 5. Experiment 1 *c* by Vowel, Group, & Class. Positive values of *c* indicate a bias towards responding e.g. HUT, with negative values indicating a bias towards responding e.g. HURT.

Table 3
Experiment 1 predictors and their levels, with baseline levels in italics. These predictors apply to both *d'* and *c* analyses.

Predictor	Factor levels
Listener Group	*Glasgow* Intermediate Cambridge
Speaker Class	*Middle Class (MC)* Working Class (WC)

ference between *bud/bird* words produced by the MC speakers (a relatively high Mean *d'* of 3.270 across all listener groups), but less so for words produced by the WC speakers (Mean *d'* = 1.752).

Table 4 shows that, when compared to the Glasgow:MC baseline for the interaction model, *Glasgow* listeners were less sensitive to difference when responding to the WC stimuli ($\beta = -0.707$, $t > \pm 2$, $p < 0.001$). There were no significant differences between the listener groups for MC *d'* stimuli, as $t < \pm 2$ ($p > 0.05$) for both Glasgow∼Intermediate and Glasgow∼Cambridge comparisons, and a similar model with Intermediate:MC set as the baseline (not shown here), showed no difference for the Intermediate∼Cambridge comparison ($t = 1.021$, $p = 0.301$).

To assess the difference in *d'* between responses to WC stimuli by *Intermediate* and *Cambridge* listeners, an LMER model with a relevelled baseline of Intermediate:WC (not shown here), showed that, compared with the Intermediate listeners, the Cambridge listeners were less sensitive to differences in minimal pairs such as *bud/bird* as produced by the WC speakers (β = -0.8311, t = -4.112, p $<$ 0.001). Another model with the baseline relevelled to Glasgow:WC showed that the Glasgow listeners were significantly more sensitive to difference than both the Intermediate listeners (β = -1.101, t = -5.515, p $<$ 0.001) and the Cambridge listeners (β = -1.932, t = -9.560, p $<$ 0.001).

2.6.2. Response bias c

In this analysis, more positive values of *c* represent a bias towards listeners responding that they heard no /r/ in the stimuli, with negative values indicating a bias towards hearing an /r/. Fig. 7 shows that there is generally lower bias in listener responses to the MC stimuli, but more variety in responses to WC stimuli.

Looking now at Table 5, the Intermediate listener group is more biased – compared to the baseline of Glasgow:MC – towards responding that they heard /r/ in the stimuli (β = -0.255, t $>\pm$ 2, p $<$ 0.05). There were no other differences between listener groups for the MC stimuli, including in a relevelled model allowing a comparison between the Intermediate and Cambridge groups.

Table 5 shows that, when compared to the Glasgow:MC baseline for the interaction model, *Cambridge* listeners were more biased to respond that they heard no /r/ in the WC stimuli (β = 0.687, t $>\pm$ 2, p $<$ 0.001) – in other words, they were more likely to respond e.g. HUT, regardless of whether the WC speaker intended to produce ‘hut’ or ‘hurt’. One result which falls just short of statistical significance (t = 1.716, p $>$ 0.05) is the comparison between the Glaswegian group’s bias when responding to MC stimuli, and to WC stimuli. This is represented as the difference between the two blue boxes on Fig. 7. Despite its non-significance, this result is worthy of note because it indicates that even Glaswegian listeners might be processing the contrast between /r/ words and non-/r/ words differently, depending on the sociolect in which the forms appear.

To assess the difference in *c* between responses to WC stimuli by *Intermediate* and *Cambridge* listeners, another LMER interaction model with a relevelled baseline of Intermediate:WC (not shown here), showed that the Cambridge listeners were more biased than the Intermediate listeners towards reporting that a word was HUT than HURT (β = 0.921, t = 7.195, p $<$ 0.001). A further model with the baseline relevelled to Glasgow:WC showed that the Intermediate listeners were significantly more biased to report hearing /r/ in the WC stimuli than the Glasgow listeners (β = -0.354, t = -2.798, p = 0.006), and the Cambridge listeners were significantly more likely to report hearing stimuli without/r/ than the Glasgow listeners (β = 0.567, t = 4.432, p $<$ 0.001).

2.7. Experiment 1 discussion

The research question for Experiment 1 was:

‘What is the role of experience in the perception of coda /r/ versus its absence in Glaswegian English?’

The results from this experiment showed a very clear effect of listener experience on the perception of derhoticised /r/, in that the Glasgow listeners, who had the most experience with hearing the Glaswegian accent, were much more sensitive to subtle differences in fine phonetic detail (with a high *d’* sensitivity value), and they also showed the least response bias (with *c* closest to 0), revealing that they were highly attuned to the phonemic categories intended by the working class speakers in the experiment.

In contrast, the Cambridge listeners’ low *d’* for WC stimuli means that they were much less sensitive to difference between *hut* and *hurt* words, and their high positive *c* for the same stimuli indicates that they showed a large bias towards hearing *hut* words, even when the speaker’s intention was to produce a word *with* /r/. The low sensitivity result shows that these listeners’ lack of experience with working class Glaswegian speech severely affects their ability to interpret fine phonetic cues to a phonological distinction, and the strong bias towards *hut* suggests that, in the absence of knowledge about the variation of such phonetic detail (due to their inability to distinguish *hut* from *hurt*), they tend towards organising stimuli into categories which are known to them. In other words, the acoustic similarity of the derhoticised *hurt* words to plain-vowel *hut* words in the WC Glaswegian speaker’s productions encourages these unfamiliar listeners to categorise them as plain-vowel *hut* words. This clearly means that not only is derhoticised /r/ very similar to back vowels acoustically (see Stuart-Smith, 2007; Lennon et al., 2015), but also perceptually, for listeners who are unfamiliar with the feature. Theoretical positions such as exemplar models and Bayesian inference would likely state that this patterning shows a clear benefit of experience for the Glasgow listeners, due to a vastly increased number of exemplars of derhoticised /r/, compared with the Cambridge listeners. This is almost certainly linked to the increased contextual and situational knowledge of the Glaswegian listeners.

A reviewer points out the important consideration that since both the WC *hut* and *hurt* stimuli are similar to the Cambridge listeners’ own representations for the *hut* category in SSBE, this might also exert an influence on their pattern of responses, and thus their *d’* and *c* values. Indeed, the canonical forms for the MC Glaswegian, WC Glaswegian, and Cambridge *hut* words are broadly the same (in vowel quality, if not in other regards such as /t/-aspiration), as they are all [hʌt], whereas the minimal pair counterpart *hurt* words are [hɚt], [hʌˤt], and [hɜːt], respectively. Since the phonemic categories we learn in childhood influence what we are exposed to later in life (Best, 1994; Flege, 1995; Tomé Lourido and Evans, 2019), the high *c* values of the Cambridge listeners could be evidence that they are more biased to report hearing *hut* because this form is already part of their inventory.

The Intermediate listener group, representing English listeners who had lived in Glasgow for around three years, showed a more interesting pattern in Experiment 1. As would be expected, their sensitivity to difference between *hut* and *hurt* WC words was between those of the Glasgow and Cambridge listener groups: their mean *d’* was at a level in between the other two groups (see Fig. 6), and it was significantly distinct from both groups (see Table 4). However, their response bias *c* had ‘overshot’ the ideal, native-like pattern that the Glasgow listeners displayed (which was little or no bias), and was now showing an effect of ‘perceptual hypercorrection’, where the listeners were over-reporting the presence of /r/ in the ambigu-

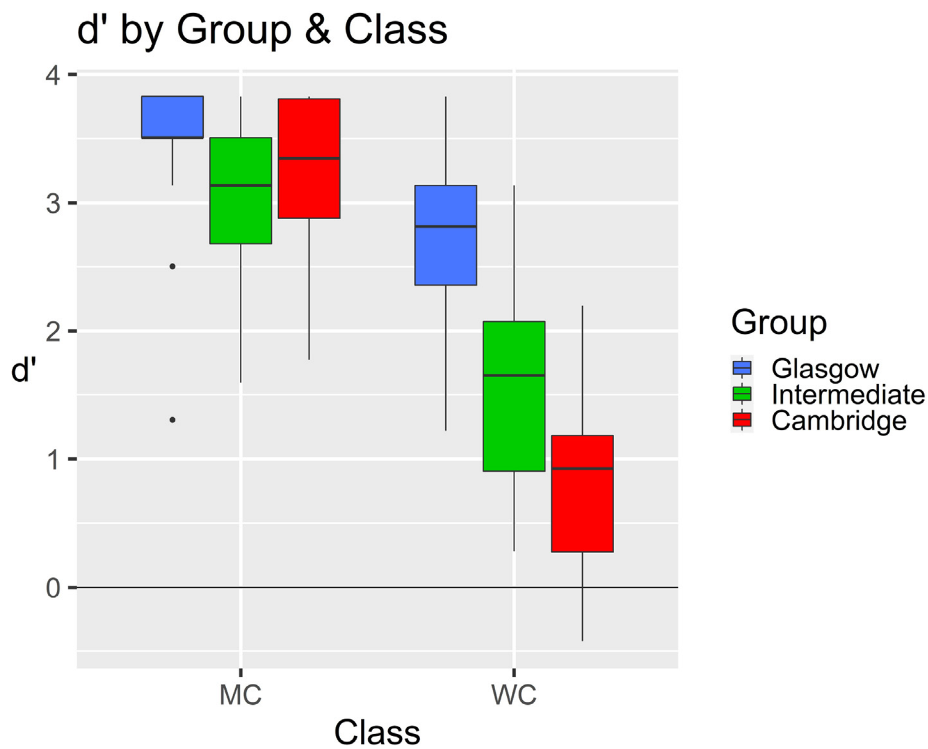


Fig. 6. Experiment 1 *d'* by Group & Class.

Table 4
Interaction model for *d'* in Experiment 1, with *Glasgow:MC* set as the reference level. Based on 124 values of *d'*, averaged across all responses to stimuli, but within factor and within subject. Higher values of *d'* indicate higher sensitivity to stimulus difference between minimal pairs in 2AFC task. Model includes random intercept of subject (s.d. = 0.419). t-values above ± 2 are indicative of a significant effect when compared to the baseline factor level and are highlighted in bold, as are their accompanying p-values.

Predictors (baseline: *Glasgow:MC*)	Estimate	Std. error	t-value	p	Mean *d'*
Listener Group					3.456
Intermediate	−0.382	0.200	−1.913	0.059	3.074
Cambridge	−0.176	0.202	−0.869	0.387	3.280
Speaker Class					3.456
Working Class (WC)	−0.707	0.152	**−4.647**	**<0.001**	2.749
*Group*Class*					3.456
Intermediate:WC	−0.719	0.215	**−3.342**	**0.001**	1.648
Cambridge:WC	−1.757	0.218	**−8.061**	**<0.001**	0.816
(intercept)	3.456	0.141	**24.481**	**<0.001**	

ous stimuli. This suggests that the Intermediate listeners had accrued knowledge about the existence of derhoticised /r/ during their time in Glasgow, learning that it is a phonetic strategy used by working class speakers to signify the presence of /r/. They may have been over-influenced by this relatively recently learned knowledge about a fairly unusual phonetic feature, and applied it to linguistic environments where they knew it could appear. On the left of Fig. 7, we can see that there is a significant effect where the Intermediate listeners respond with a bias towards hearing /r/ in the MC stimuli (as well as the WC stimuli), compared with the Glasgow listeners hearing the same stimuli ($t = -2.019$, $p < 0.05$). This could be because their hypercorrected response bias – motivated by their experience with derhoticised /r/ in the WC stimuli – also influences their perception of the MC *hut*-type stimuli, which are presented in the same experimental blocks as the WC stimuli (it is unlikely that the their perception of the strongly-rhotic MC *hurt*-type stimuli is affected in this way). Alternatively, it could even suggest that the Intermediate listeners are biased towards perceiving more /r/ realisations in Scottish speakers overall, as Scottish speech is stereotypically rhotic. This seems less likely than the potential for an experimental effect, but it is not possible to determine which explanation is the most likely with the current data.

An interesting observation can be made concerning the Glasgow listeners' high performance in this task, which shows high sensitivity and low bias in their responses to the WC stimuli. Despite the Glaswegian listeners almost all displaying strongly-rhotic productions in their own speech,[4] their perception of derhoticised /r/ (which they do not themselves produce)

[4] This informal observation was made during testing, with participants recruited from the mostly middle class cohort attending the University of Glasgow. However, since socioeconomic data and speech recordings of participants were not taken, this pattern cannot be formally tested with the current data.

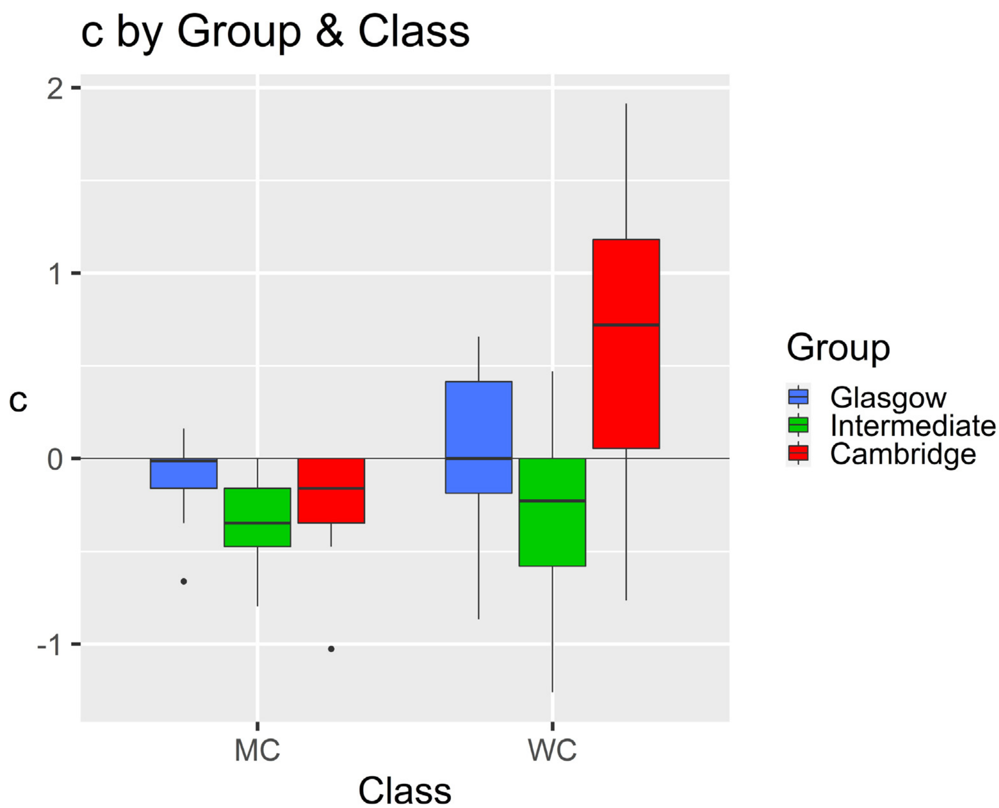


Fig. 7. Experiment 1 *c* by Group & Class. Positive values of *c* indicate a bias towards responding e.g. HUT, with negative values indicating a bias towards responding e.g. HURT.

Table 5
Interaction model for *c*, with *Glasgow:MC* set as the reference level. Based on 124 values of *c*, averaged across all responses to stimuli, but within factor and within subject. Positive values of c indicate a bias towards responding e.g. HUT, with negative values indicating a bias towards responding e.g. HURT. Model includes random intercept of Subject (s.d. = 0.263). t-values above ± 2 are indicative of a significant effect when compared to the baseline factor level and are highlighted in bold, as are their accompanying p-values.

Predictors (baseline: *Glasgow:MC*)	Estimate	Std. error	t-value	p	Mean *c*
Listener Group					−0.091
Intermediate	−0.255	0.126	**−2.019**	**0.046**	−0.346
Cambridge	−0.120	0.128	−0.934	0.353	−0.210
Speaker Class					−0.091
Working Class (WC)	0.166	0.097	1.716	0.091	0.076
*Group*Class*					−0.091
Intermediate:WC	−0.098	0.137	−0.718	0.476	−0.278
Cambridge:WC	0.687	0.139	**4.946**	**<0.001**	0.643
(intercept)	−0.091	0.089	−1.013	0.314	

is nearly at the same level as that of the contrast between the MC stimuli. Since their performance is notably better than that of either group of listeners raised in England, one possible interpretation is that the concept of the ‘fluent listener’ (Sumner and Samuel, 2009) may be in play, meaning that the Glasgow listeners could simply be better at perceiving different types of /r/, owing to the fact that there may be a higher degree of /r/ allophony in the Scottish Central Belt area (e.g. Lawson et al., 2011b; Lawson et al., 2014; Dickson and Hall-Lew, 2017; Jauriberry, 2021) than in England. Of course, a simpler explanation could be that the Glasgow listeners have had much more exposure to derhoticised /r/ (around twenty years, compared to three for the Intermediate group), which is reflected in the pattern of sensitivity seen in Fig. 6, showing that *d’* is correlated with exposure.

Despite its non-significance ($t = 1.716$, $p > 0.05$), one of the response bias results in Experiment 1 warrants further discussion. In Fig. 7, the difference between the blue boxes shows that Glaswegian listeners were very slightly biased towards reporting hearing /r/-less words when hearing the working class speaker. This trend was unsurprising – despite the Glaswegian listeners’ high sensitivity to difference between the working class pairs (as shown by their high *d’* results), the vowel-like formant structure of *hut* and *hurt* words produced by the working class speakers made it likely that there would be a slight bias towards them reporting the absence of an /r/. However, they also displayed a trend of being slightly biased to report that the MC stimuli were /r/-ful (mean $c = -0.091$), regardless of whether the stimulus did in fact contain an /r/: in other words, some responses by Glasgow listeners indicated that they perceived e.g. *hut* as *hurt*. This trend is in line with the bias results of the other listeners: both the Cambridge and Intermediate listener groups were biased towards reporting more /r/-ful words in the middle class stimuli. However, it is most surprising in the Glaswegian listeners, as it was predicted that their increased experience with the accents in question would help them to show no bias at all in the 'easy' distinction between *hut* and *hurt*, the latter of which has a highly-rhotic acoustic quality, thus increasing the auditory difference in their decision about contrastive rhoticity in the experiment.

Despite the fact that the bias effect in the Glasgow listener group is a non-significant trend, if future work on this did indicate a more robust result in this direction it would have important implications for the wider literature on Scottish rhoticity. Recall Figs. 1,2 from Section 1.1, which describe the suggestion in Lawson et al. (2011b) and Lawson et al. (2014) of potential correlation between auditory judgements of /r/-realisation and the same speakers' articulatory configurations, in data collected across Scotland's Central Belt. Experiment 1 in the present work is the first known perceptual study which has tested this in the wider population, and the bias pattern of the Glaswegian listeners in Experiment 1, as described above, may indicate the existence of a split between how middle class and working class /r/ variants are perceived. In other words, it could be the case that Glaswegian listeners know about the difference between these /r/ variants, and judge the speakers of the two sociolects differently based on the indexical information they have inferred about them. This (non-significant) trend could be what makes the case for listeners' use of indexical information. However, the fact that Glaswegian listeners appear to know what a derhoticised /r/ is – and accept it as an /r/ – does not necessarily mean that they have made any inferences about the social status of the speaker, and in order to address this issue more directly, the social element needs to be tested further in separate perceptual experiments.

The co-dependent nature of social and phonological information may be accounted for by using the kind of representations put forward in exemplar theory. Since the response bias for Glaswegian listeners is *towards* /r/ for the strongly rhotic middle class speakers, but *away from* /r/ for the weakly rhotic working class speakers (Fig. 7), this means they are classifying the middle class speakers as being more /r/-ful in general, even in their *hut* words, and the opposite for the working class speakers, as noted above. If the Glasgow listeners are altering their judgements of individual *hut* and *hurt* tokens based on the identity of the speaker who produces them, this is evidence that they are using the speaker's identity to shape their expectation of which /r/ variant they are likely to hear. This mirrors the 'integrated talker and phoneme processing' that Mullennix and Pisoni (1990) found in their influential experiment, which was one of the catalysts for the growth in popularity of exemplar theory.

Crucially though, the Glasgow listeners still primarily treat both the middle class [ɚ] and the working class [ʌˤ] as /r/, to a much greater degree than either of the English listener groups (Intermediate and Cambridge), underlining the benefit of increased exposure. Moreover, the Glaswegian listeners have acquired knowledge of the system in an entirely different way. One issue which has not been raised yet in this paper is the possibility of the speaker's 'own accent' having an effect on their perception. In general terms it is likely that a listener who hears the same accent as their own will benefit from this fact, but it is undoubtedly very difficult to tease this apart from their accent exposure, that is, the accents they hear around them.

2.8. Experiment 1 summary

This analysis shows a clear effect of accent experience on a listener's ability to distinguish between perceptually ambiguous words, which are known from Lennon et al. (2015) to be acoustically very similar. If the response bias and sensitivity results are taken together, it may be deduced that the Intermediate listeners have improved their perception of derhoticised /r/, in that they are much better at discerning differences between the words than the Cambridge group. However, their bias towards hearing /r/ in the /ʌ/ tokens – compared with the Glaswegian listeners – indicates that they are still a long way from being native-like in their perception of the feature, suggesting that much more than three years' casual (and possibly infrequent) exposure to an unfamiliar phonetic feature is needed for improvement. This also raises the question of what happens when inexperienced listeners (like those in the Cambridge listener group) start to learn fine phonetic detail. How long might it take to reach the stage of proficiency that the Intermediate listeners in Experiment 1 have achieved? This question is addressed in the next experiment.

3. Experiment 2

The purpose of Experiment 1 was to assess the effect of long-term familiarity with /r/ in Glasgow. Experiment 2 now adds to this by examining the role of short-term learning in a listener's perception of unfamiliar phonetic detail, in order to address *RQ2: 'How does listener experience relate to the effect of exposure on the perception of a non-standard phonetic contrast?'*. We now focus on the perception of derhoticised /r/, which was generally produced by WC speakers, as Experiment 1 found evidence that it was the most challenging variant for listeners to process. Where Experiment 1 tested the effect of long-term listener experience, Experiment 2 investigates what happens when listeners begin to learn the distinction between these words. In other words, Experiment 2 tests participants with different levels of experience of a phonetic contrast on their ability to learn fine phonetic detail over a short period of time.

3.1. Design

In order to achieve this aim, Experiment 2 uses a Pretest-Exposure-Posttest design, with 2AFC tasks as the Pretest and Posttest. Any differences in the patterns of results between Pretest and Posttest would suggest that the Exposure section has caused a change in the listeners' perception of derhoticised /r/, which might indicate the early stages of learning this phonetic segment.[5] The Exposure section took the form

[5] Experiment 2 originally included experimental and control conditions, with F2 and F3 in the V(+/r/) sequence acoustically manipulated to remove differences between minimal pairs in the control stimuli, as it was thought that this might provide insight into whether listeners were indeed learning from the phonetic detail in the derhoticised /r/ and not some other feature of the speaker's production in the Exposure story. However, since early statistical analyses showed no significant differences between listener responses in the two conditions – either as a main effect or in interaction with any other factor – the control stimuli (and therefore the factor of condition) were excluded from further analysis.

of a 6-min long story concerning a man's walk through the countryside, read by the same speaker as in the 2AFC stimuli. Because the story contained minimal pairs with and without derhoticised /r/ (e.g. *thud/third*), this provided listeners with the opportunity to learn the contrast between the speaker's production of /r/ and the plain vowel, and the story ensured that the ambiguous words were surrounded by the linguistic context necessary to learn the identity of the speaker's intended phoneme in each case.

3.2. Materials

The stimuli in Experiment 2 were all produced by one of the WC males from Experiment 1, who was 28 years old by the time the stimuli for Experiment 2 were recorded. These stimuli, as well as the Exposure story, were recorded in a new session for Experiment 2, under the same recording conditions as for Experiment 1. Given the ceiling effect for /i/ stimuli (discussed in Section 2.5), the target stimuli in Experiment 2 were all /ʌ/ words, shown in Table 6. This table also shows the distractor stimuli, which were chosen because – like *hut* [hʌt] and derhoticised *hurt* [hʌˤt] – they included distinctions that are similarly close phonetically in Glaswegian English, thus were deemed to be effective distractors. These pairs were: /b∼p/ (e.g. *Ben/pen*), /e∼i/ (*fade/feed*) and /o∼ɔ/ (*code/cod*).

3.3. Procedure

As in Experiment 1, participants in both of the 2AFC tasks in Experiment 2 were asked to choose which option out of two on-screen alternatives that they heard in the stimuli. In the Pretest, listeners heard 96 randomised stimuli – including 72 distractors – where the 24 targets were isolated words, as in Experiment 1. Otherwise, the procedure for the 2AFC Pretest and Posttest tasks was identical to that of Experiment 1, described in Section 2.3. After the Pretest task, participants were played the Exposure story, in which listeners were exposed to twelve minimal pair target contrasts (24 words), randomly placed throughout the story. To ensure attention was paid, they were asked to keep a total of the number of times an animal was mentioned by the speaker. Following the Exposure story, they completed the Posttest, which was identical to the Pretest, except it was presented in a different randomised order.

3.4. Participants

As in Experiment 1, native monolingual English-speaking listeners were recruited from both the University of Glasgow (Glasgow and Intermediate groups) and the University of Cambridge (Cambridge group), and were paid a small fee for participation. Of the 128 participants who took part in the experiment, half (n = 64) were in the experimental condition, so only their responses will be analysed in this paper – see Footnote 5 for the reason behind the exclusion of the control condition. None reported any hearing or language difficulties, and none had taken part in Experiment 1:

- **Glasgow** (n = 21): Raised in Glasgow, living in Glasgow;
- **Intermediate** (n = 22): Raised in England, living in Glasgow (mean residence = 3.5 years);
- **Cambridge** (n = 21): Raised in South East England (little/no experience of Glaswegian).

Table 6
Minimal pairs used in Experiment 2. Two tokens of each word in each pair (both target and distractor) appeared in the Pretest and Posttest, resulting in 96 words in each test.

Target stimuli	Distractor stimuli
bud / bird	*Ben / pen*
bun / burn	*bet / pet*
hut / hurt	*big / pig*
shut / shirt	*bin / pin*
thud / third	*bit / pit*
tonne / turn	*bunch / punch*
	coast / cost
	code / cod
	fade / feed
	fate / feet
	goat / got
	hate / heat
	hope / hop
	mane / mean
	mate / meat
	shape / sheep
	shown / shone
	soak / sock

3.5. Statistical analysis

The same statistical modelling strategy from Experiment 1 was followed in Experiment 2, with the *lme4* R package used to build fully-saturated models, and *LmerTest*'s *step()* function used to eliminate non-significant effects. The remaining predictors – with treatment-coded variables – for both the *d'* and *c* models are shown in Table 7.

3.6. Results

3.6.1. Sensitivity d'

Fig. 8 shows data for the non-significant interaction of Test by Group for *d'*, and shows a significant main effect of listeners overall improving their sensitivity to stimulus difference between Pretest and Posttest (β = 0.227, t $>\pm$ 2, p $<$ 0.01). The data also shows a significant main effect of Listener Group on the responses to stimuli in Experiment 2, across both Pretest and Posttest tasks. The Glasgow group clearly has the highest value of *d'* in this experiment overall, followed by the Intermediate group, then Cambridge. Table 8 shows significant differences between the Glasgow (the model's baseline) and Intermediate groups (β = -1.493, t $>\pm$ 2, p $<$ 0.001), as well as between Glasgow and Cambridge groups (β = $-$2.064, t $>\pm$ 2, p $<$ 0.001). Another LMER model with a relevelled baseline (not shown here) showed that the difference between Intermediate and Cambridge groups in their sensitivity to

Table 7
Experiment 2 predictors and their levels. These predictors apply to both *d'* and *c* analysis, with *Posttest* (in italics) the Test baseline in both. In the *d'* model, the Listener Group baseline is *Glasgow*, and in the *c* model, it is *Cambridge*.

Predictor	**Factor levels**
Listener Group	Glasgow
	Intermediate
	Cambridge
Test	Pretest
	Posttest

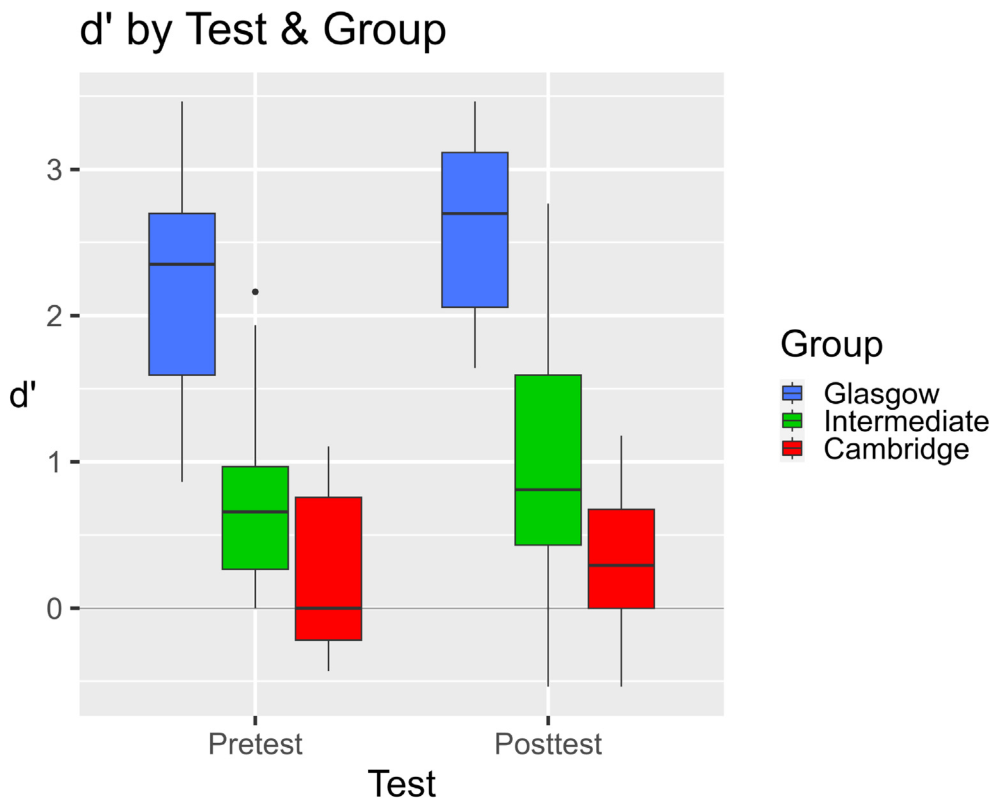


Fig. 8. Experiment 2 *d'* by Test and Group.

Table 8
Final simple model for *d'*, with *Glasgow* set as the reference level. Based on 128 values of *d'*, averaged across all responses to stimuli, but within factor and within subject. Higher values of *d'* indicate higher sensitivity to stimulus difference between minimal pairs in 2AFC task. Model includes random intercept of Subject (s.d. = 0.469). t-values above ± 2 are indicative of a significant effect when compared to the baseline factor level and are highlighted in bold, as are their accompanying p-values.

Predictors	Estimate	Std. error	t-value	p	Mean *d'*
List. Group (baseline: *Glasgow*)					2.351
Intermediate	−1.493	0.170	**−8.760**	<**0.001**	0.858
Cambridge	−2.064	0.172	**−11.967**	<**0.001**	0.287
Test (baseline: *Pretest*)					1.047
Posttest	0.227	0.076	**2.993**	**0.004**	1.274
(intercept)	2.238	0.128	**17.526**	<**0.001**	

Table 9
Interaction model for *d'*, in which the interaction terms were non-significant. *Glasgow:Pretest* is set as the reference level. Based on 128 values of *d'*, averaged across all responses to stimuli, but within factor and within subject. Model includes random intercept of Subject (s.d. = 0.470). t-values above ± 2 are indicative of a significant effect when compared to the baseline factor level and are highlighted in bold, as are their accompanying p-values.

Predictors (baseline: *Glasgow:Pretest*)	Estimate	Std. error	t-value	p	Mean *d'*
Listener Group					2.168
Intermediate	−1.430	0.194	**−7.381**	<**0.001**	0.738
Cambridge	−1.916	0.196	**−9.778**	<**0.001**	0.251
Test					2.168
Posttest	0.367	0.132	**2.785**	**0.007**	2.534
*Group*Test*					2.168
Intermediate:Posttest	−0.127	0.184	−0.687	0.495	0.978
Cambridge:Posttest	−0.295	0.186	−1.581	0.119	0.324
(intercept)	2.168	0.139	**15.642**	<**0.001**	

stimulus difference was significant, with the Intermediate group having more sensitivity to difference (β = 0.486, t = 2.510, p = 0.014). Table 9 shows the Group*Test interaction model for the data in Fig. 8, and since the model baseline is set as Glasgow:Pretest, we can see a significant improvement in the Glasgow listeners' sensitivity to stimulus difference between the Pretest and Posttest tasks (β = 0.367, t $>\pm$ 2, p $<$ 0.01). However, the lack of an interaction effect between Group and Test indicates that the effect of Test was not significantly different between listener groups, meaning that all listener groups improved in their sensitivity to a similar extent between Pretest and Posttest.

3.6.2. Response bias c

Fig. 9 shows data for the interaction of Test by Group for response bias *c*, with the accompanying Table 10 showing the model. This allows us to see that there was a significant change in the Cambridge listeners' response bias towards reporting more /r/ between the Pretest and Posttest tasks (β = $-$0.368, t $>\pm$ 2, p $<$ 0.001). Similar models (not shown here) with Intermediate:Pretest and Glasgow:Pretest set as their baselines showed a significant change in the Intermediate listeners' response bias towards reporting more /r/ between the Pretest and Posttest tasks (β = $-$0.308, t = $-$3.358, p = 0.0014), but no change for the Glasgow listeners (t = 0.029, p = 0.977). It is this difference between the groups which results in the interaction's statistical significance. In summary, both Intermediate and Cambridge listener groups showed a greater bias towards reporting e.g. HURT in Posttest relative to Pretest, whereas the Glasgow group showed no such change.

Another model without the interaction term (not shown here) showed only a significant main effect of Test, such that there was a change operating across all groups in response bias *c*

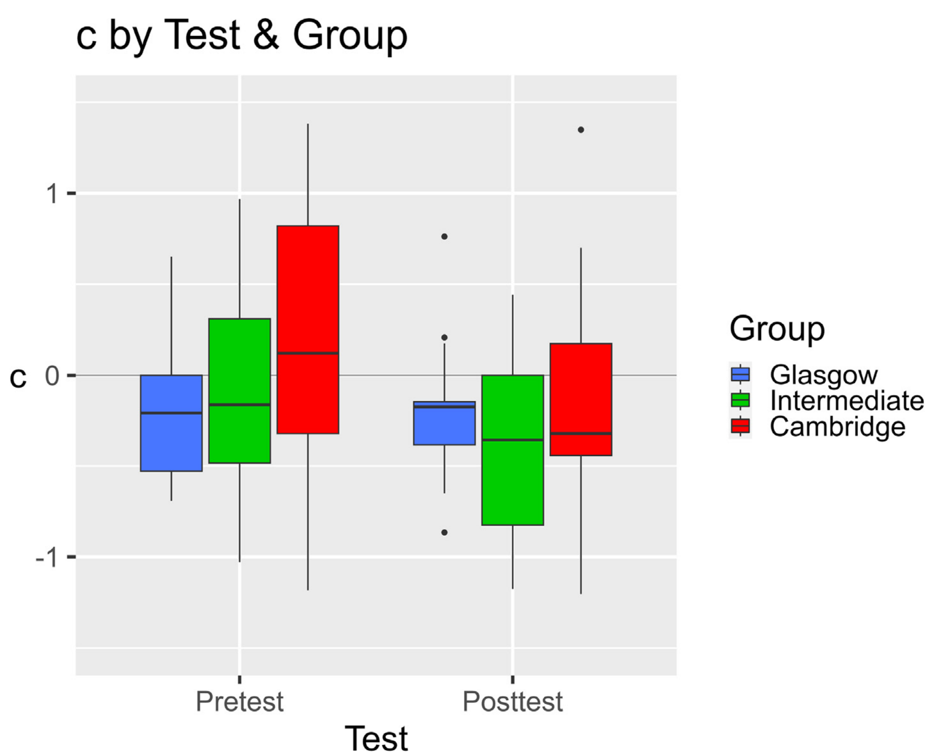


Fig. 9. Experiment 2 *c* by Test and Group.

Table 10
Interaction model for *c*, with *Cambridge:Pretest* set as the reference level. Based on 128 values of *c*, averaged across all responses to stimuli, but within factor and within subject. Positive values of c indicate a bias towards responding e.g. HUT, with negative values indicating a bias towards responding e.g. HURT. Model includes random intercept of Subject (s.d. = 0.435). t-values above ± 2 are indicative of a significant effect when compared to the baseline factor level and are highlighted in bold, as are their accompanying p-values.

Predictors (baseline: *Cambridge:Pretest*)	Estimate	Std. error	t-value	p	Mean *c*
Listener Group					0.196
Glasgow	$-$0.412	0.164	**$-$2.512**	**0.014**	$-$0.216
Intermediate	$-$0.261	0.162	$-$1.609	0.111	$-$0.065
Test					0.196
Posttest	$-$0.368	0.094	**$-$3.919**	$<$**0.001**	$-$0.173
*Group*Test*					0.196
Glasgow:Posttest	0.371	0.133	**2.791**	**0.007**	$-$0.213
Intermediate:Posttest	0.060	0.131	0.457	0.650	$-$0.373
(intercept)	0.196	0.116	1.690	0.095	

towards reporting /r/ in the stimuli following exposure to the story ($\beta = -0.226$, t = -3.981, $p < 0.001$). There was no difference between listener groups in their overall bias in this model.

3.7. Experiment 2 discussion

Experiment 1 considered the effect of long-term experience of the fine phonetic contrast in minimal pairs with or without derhoticised /r/. The focus of Experiment 2 shifted to short-term learning, testing what happens when listeners who are potentially unfamiliar with the Glaswegian linguistic environment have the opportunity to experience this variant. The research question for Experiment 2 was:

'How does listener experience relate to the effect of exposure on the perception of a non-standard phonetic contrast?'

The analysis once again showed an effect of listener experience on the perception of derhoticised /r/, comparing the WC boxes on the right hand side of Fig. 6 with all boxes in Fig. 8 (recall that Experiment 2 only tested WC stimuli), with the Glasgow listeners again being the most sensitive of the three groups to stimulus difference. The short-term adaptation element of this experiment showed an overall increase in the listeners' sensitivity to stimulus difference, with a higher *d'* in Posttest than in Pretest, indicating improvement in ability to discriminate stimuli following the Exposure story. Response bias also had an overall shift, with listeners overall being more biased towards reporting that stimuli had an /r/ in Posttest than in Pretest.

Looking at Fig. 9 and Table 10, response bias for Cambridge listeners shifted – following short-term learning – from *hut* to *hurt*, thus closely matching the long-term bias of the Intermediate listeners. This suggests a very fast adaptation in the listeners' perceptual systems, and appears to signal a rapid change in their vowel and /r/ categorisation criterion for this speaker.[6] Presumably they would then be able to expand this newly altered categorisation criterion to other speakers of working class Glaswegian that they encountered, but it would be difficult to predict how long this effect might last. Experiments by Kraljic and Samuel (2005), Eisner and McQueen (2006), and others, have shown that this type of knowledge is retained after 25 min, and 12 h, respectively.

Another useful comparison to make is between the relative lack of bias of the Glasgow listeners to either MC or WC stimuli in Experiment 1, seen in Fig. 7, and the Glasgow listeners' slight bias towards reporting hearing an /r/, seen in Fig. 9. This difference between the Glasgow groups in each experiment is almost certainly related to the difference in the experimental blocks, with Experiment 2 only presenting a single working class speaker to the participants. The Glasgow group's pattern in Experiment 2 is in fact similar to that of the Intermediate groups' long-term familiarity with the WC stimuli in both experiments, and the Cambridge group's bias shift towards hearing more /r/s in Experiment 2 (Fig. 9). This means that, in addition to both Intermediate groups' theorised 'perceptual hypercorrection', discussed in Section 2.7, and the Cambridge group seeming to show the same pattern after short-term exposure, the Glasgow listeners may also be displaying the same perceptual hypercorrection towards hearing /r/ in the WC stimuli in Experiment 2.

However, the fact that the Glasgow listeners show no bias when responding to the WC stimuli in Experiment 1 (Fig. 7), suggests that the context of hearing both the middle class and working class /r/ variants in the same session may have assisted them in their perception of derhoticised /r/. If this is the case, the fact that the Glasgow listeners in Experiment 1 had access to a greater range of indexical information due to their increased experience with the Glaswegian linguistic environment (across accents *and* speakers) actually assisted their perception of fine phonetic detail in derhoticised /r/. One interpretation of this is that it could be evidence for the benefit of indexical information in speech processing. This might lend support to exemplar-based theories of speech perception, proponents of which generally claim that listeners compare incoming phonetic and other linguistic information to stored categories, and goodness-of-fit judgements are made for each new exemplar. It is possible that the existence in the same task of lots of indexical information allowed the Glasgow listeners in Experiment 1 to perform goodness-of-fit comparisons armed with much more information in their immediate and very short-term memory than the listeners in Experiment 2 had, as the latter group only heard one speaker and accent, and therefore did not have the benefit of a larger inventory of indexical information. Furthermore, Glasgow listeners are the only listener group of the three who demonstrate this apparent benefit from indexical information, so it is likely that the native listeners are able to put their much larger repository of exemplars of Glaswegian /r/ (of all types, strong and weak) to good use when categorising new linguistic data.

One possible mechanism of exemplar-based speech perception could be that listeners build up their collection of exemplars for a word (or vowel, or consonant, etc.), say, *yesterday*, into a most-likely pattern. This could be how listeners construct categories for words or phonemes, over time. The listener also can apply knowledge of things like speech rate, interference due to noisy conditions, and other conditioning factors, to their pattern recognition process. This means that they would be acting like the 'ideal adapter' of Kleinschmidt and Jaeger's (2015) Bayesian model of speech perception. This allows for the possibility that, if a listener had for some reason only ever heard *yesterday* as *yeshay*, a reduced form which usually arises due to fast speech (see Ernestus, 2014), they might be unaware that it is indeed a reduced form, and may be surprised when they hear the canonical form of *yesterday*. This surprise is perhaps evident in the Cambridge listeners' rapid change in response bias from Pretest to Posttest in Experiment 2 (Fig. 9). Perhaps more realistically, if the listener had only heard a certain word (or vowel, or consonant, etc.) produced in a certain way by listeners speaking with a particular accent, then they heard a different realisation of that word, they might experience difficulty in recognising it as intended by the speaker. The listener would then have to activate a number of conditioning factors that they may estimate to be in play, that may be affecting their recognition of the word. Further speech perception experiments could go some way to identifying which components of any processing costs (e.g. increased response times, which were not measured here) can be attrib-

[6] Interestingly, the response bias for Intermediate listeners shifted from a bias for *hurt* towards an even greater bias for *hurt*, possibly suggesting that the Exposure story boosted their knowledge about the existence of derhoticised /r/, which resulted in a tendency to perceptually hypercorrect to an even greater degree.

uted to particular difficulties a listener might experience with a linguistic phenomenon.

Of course, the conditioning factors that are available to a listener are only those which they have had the need to apply in the past – in other words, if a listener has experience of travelling to many different locations where speakers have different accents, then they have a lot of experience with the fact that certain pronunciations vary due to location. Moving to a different accent area may then alter their group of perceived exemplars for that word (though see Evans and Iverson, 2007, where this does not reliably happen), with the category perhaps becoming expanded for a while, then changing to the category found in the new location. They would presumably still have the old exemplars in their memory, so they would likely still be able to easily adapt to hearing that pronunciation.

3.8. Experiment 2 summary

The clearest result from Experiment 2 is that the listeners raised in Glasgow were overall the best performing group, which supports the overall pattern of long term learning seen in Experiment 1. The Intermediate and Cambridge listeners, who had much less experience with Glaswegian, showed some intriguing patterns, with the Cambridge listeners appearing to show the beginnings of perceptual change in their swing in response bias, thus matching the long-term experience pattern of the Intermediate listeners. Interestingly, the *d'* results in Fig. 8 are remarkably similar to the *d'* results for the WC *hut/hurt*-type stimuli in Experiment 1 (Fig. 6), suggesting that both experiments are consistent in their finding of an effect of long-term familiarity. Because of the differences between the two experiments it is not meaningful to conduct any statistical analysis to test this similarity, so this is purely an impressionistic observation, however, it suggests that the effect of long-term experience on derhoticised /r/ could be relatively stable across different listener populations with similar demographics.

However, the result with the most relevance to *RQ2* – how experience interacts with exposure – is that all listener groups showed a similar degree of improvement in their sensitivity to stimulus difference, possibly indicating a benefit for short-term exposure to a phonetic contrast that does not depend on the listener's long-term experience with it. It is also possible that this indicates an improvement in phonetic perception associated with a short-term increase in familiarity with the speaker who produced the stimuli, but without further controlled experiments it is not possible to determine whether this is the case.

4. General discussion

This paper has demonstrated that the perception of one feature of fine phonetic detail in an accent of English can be a complex process, depending on both probable prior exposure of the listener to that feature outwith the laboratory, and on their ability to adapt to it quickly from limited but useful data in an experimental learning task. Experiment 1 examined the role of long-term experience with a variety of non-standard English in the ability to distinguish minimal pairs which vary across a fine phonetic contrast. Experiment 2 then investigated short-term learning, addressing the important question of what happens in the very early stages of learning a new accent feature, in other words, perceptual adaptation to the fine phonetic detail of derhoticised /r/ in Glasgow, a sociophonetic change in progress. We shall now reflect on the pattern of results from both experiments together, and consider how they relate to the theoretical positions described in the introduction.

The consistent effect of Class within each listener group in Experiment 1 suggests that variation is important for the listener. If it were the case that variation is simply stripped away as earlier abstractionist theories of speech perception suggest, converting the signal into discrete categories (e.g. McClelland and Elman, 1986; Studdert-Kennedy, 1976), or phoneme strings (Halle, 1985), then the variation presented to the listener group in Experiment 1 as a difference in speaker class might be expected to have little effect upon the results of this experiment. However, the results presented here suggest that such variation does matter, so this appears to be evidence supporting a more nuanced view of speech perception than abstractionist theories can provide. Of course, other models of speech perception besides exemplar theory (and models based upon Bayesian inference) may explain some of the results presented here, such as a benefit for experience in detecting phonetic contrast, but the notion in many exemplar studies that fine-grained detail about the talker affects perception of linguistic units does appear consistent with these results. The exemplar position put forward by Johnson (1997), suggests that listeners may be able to perform speaker normalisation, comparing the incoming signal with their stored exemplars, with accent normalisation potentially being a similar process (e.g. Nygaard and Pisoni, 1998). In Experiment 1, the middle class stimuli elicited generally better performance from the listeners than the working class stimuli, suggesting that if speaker or accent normalisation takes place it does not do so on an equal basis for all talkers.

Of course, owing to the relatively high performance for all /i/ stimuli (not formally presented here, but shown in Figs. 4,5), which have much more distinct formant structures between minimal pairs than the /ʌ/ stimulus pairs (Fig. 3), it seems likely that the primary contributor to the pattern of performance seen in Experiment 1 is the relative acoustic similarity of the working class /ʌ/ stimulus pairs, compared with the middle class /ʌ/ pairs and the /i/ pairs for both classes. Still, the patterning of results presented here may be entirely in line with the broadest interpretation of the exemplar approach, which holds that incoming exemplars are matched against the inventory of previously stored exemplars, enabling a probabilistic judgement about the phoneme's identity. The fact that the listeners in Experiment 1 are generally better at identifying the words spoken by the middle class Glaswegians is a possible indicator that they have more stored exemplars for the middle class /r/ than for the working class /r/. It is certainly the case that testing for the Glasgow and Intermediate groups took place on the campus of Glasgow University, and the Cambridge listener group may have had more opportunity to experience the middle class Glaswegian accent than the working class accent, through media exposure or other Cambridge University students who were raised in Scotland. However, in Experiment 1 this is only likely to be true for the Glasgow and (to a lesser extent) Intermediate listener groups, as the Cambridge listeners probably do not have a large enough inventory of middle class Glaswegian exemplars with which to perform an effective matching process – despite the fact that they are more likely to be familiar with middle class speech than working class speech, their familiarity with Glaswegian accents in general will be relatively low compared to that of the other listener groups. However, despite their lack of familiarity, the Cambridge listeners performed almost as well as the other groups when responding to middle class stimuli, as can be seen in both Fig. 6 for *d'* and Fig. 7 for *c*. It is of course important to note that exemplar approaches are not the only models of speech perception that can account for the changes in speaker performance in both Experiment 1 and Experiment 2, or indeed the process of speaker normalisation, as discussed above.

It is also possible to interpret these results in terms of hybrid approaches, which hold that abstract, symbolic representations exist alongside talker-specific and other indexical information (Schacter and Church, 1992). It could be the case that the Cambridge listeners in Experiment 1 are mapping the incoming exemplars of e.g. [hʌˤt] (working class *hurt*) to the category /hʌt/, which constitutes their abstract representation for the form. The fact that they also correctly categorise [hʌt] (working class *hut*) as /hʌt/ may explain their very low sensitivity to stimulus difference between WC *hut/hurt* words (Fig. 6) and and their extreme bias to classifying them as /r/-less (Fig. 7). A hybrid interpretation may also explain the result of the Cambridge listener group shifting their bias towards reporting more WC *hut/hurt* items as /hʌrt/, as seen in the change between the red Pretest and Posttest boxes in Fig. 9. If it is the case that speaker-specific detail is stored by the listener when identifying words, the Cambridge listeners (and to an extent the Intermediate listeners, also in Fig. 9) may be using the speaker-specific information they heard in the exposure story in order to refine and encode the input from this particular speaker. This may then enable the listeners to more effectively assign the incoming [hʌt] and [hʌˤt] tokens to /hʌt/ and /hʌrt/ respectively, thus improving their ability to detect the phonemic contrast and apply it to new instances from this speaker. Not only is it possible that the listeners have begun to systematically associate allophonic details with words for this speaker (Pierrehumbert, 2002), but they may also have used the exposure story to improve the process of pattern-matching between signal and memories, using the rich hierarchical structures proposed by the Polysp model (Hawkins and Smith, 2001; Hawkins, 2003; Hawkins, 2010; Smith, 2015).

Another possible theoretical stance that could be used to interpret these results is a Bayesian approach to speech perception, which suggests that the listener makes decisions regarding the identity of, for example, phonemes, based upon a combination of evidence from the speech signal and their knowledge or expectation of which phonemes they are likely to hear. Experiment 2 directly tested for the effects of short term exposure on the listeners' ability to perceive contrast in minimal pairs with plain vowels and derhoticised /r/, so it is possible to frame these results in Bayesian terms. The change in bias towards hearing more /r/ from Pretest to Posttest in both the Intermediate and Cambridge groups in Experiment 2 (Fig. 9) suggests that the listeners began to change their expectations of what the speaker intended to produce, and perhaps more specifically, that they started to add instances of derhoticised /r/, i.e. [hʌˤt] for *hurt*, to their inventory. It must be noted that no Bayesian analysis has been done in this work, but this is an avenue for future study, and, as with the exemplar interpretation above, it is also important to note that Bayesian approaches are not the only way to account for the patterning of results seen in this paper. However, it would be particularly interesting to explore this issue in terms of Kleinschmidt and colleagues' suggestion that there may be a deep connection between sociolinguistics and psycholinguistics.

5. Conclusion

This paper has presented a set of experimental results which provide insight into how listeners perceive a complex and changing speech sound, derhoticised /r/. It appears to be the case that the majority of these results are best explained by exemplar theory, as they suggest that all listener groups in Experiments 1 and 2 display a degree of plasticity in their perception of phonemic categories, and the results of Experiment 2 appear to show that category changes can be relatively rapid. This is at odds with traditionally abstractionist views, as they may not allow for such a high degree of short term variability in their (presumably fairly rigid) categories. These results may also be explained by certain hybrid models of perception, which claim that there are abstract categories at some level, but the speaker and indexical information cannot be ignored. In any case, these results are an argument against the abstractionist view that information about the talker is 'stripped away' and discarded to reveal the underlying abstract representations alone.

A final consideration is that in the experiments described in this paper, the vast majority of Glaswegian listeners spoke with middle class accents, due to the fact that recruitment took place in the University of Glasgow, as noted above. In the scope of this work this issue was unavoidable, and the same issue likely affects many perceptual studies which are conducted on university campuses. However, this, in combination with the results of Experiments 1 and 2, could promote the question for future research: *'Why do middle class Glaswegian listeners perform so well when hearing working class derhoticised /r/?'* It is possible that they have an acute awareness of the articulatory configurations and timings that are required to produce derhoticised /r/ with a tip-up gesture in Glaswegian, as that strongly correlates with pharyngealisation in the working class accents in the Central Belt of Scotland (e.g. Lawson et al., 2018), which is what they actually hear. This would need to be addressed in relation to more than one theoretical standpoint for speech perception. Of course, this could still be purely a matter of perceptual experience, rather than an effect of the speaker's own accent or knowledge of articulatory strategies, but further experiments along these lines would help to explore this in more detail.

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5.2. Prior publication

Part of the analysis described in this paper was presented at the Eighteenth International Congress of Phonetic Sciences, Glasgow (Lennon et al., 2015).

Declaration of Competing Interest

The author declares that there is no known competing financial interest or personal relationship that could have appeared to influence the work reported in this paper.

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